

Decision-making under Uncertainty and Risk

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Abstract. Decision-making is a critical process in both daily experiences and AI research that involves selecting actions among alternatives. Several studies have implemented various techniques, such as rule-based systems, to examine the consequences of decisions in their simplest form. While decision-making is essential, in matters of extreme weather events where uncertainty and high-risk outcomes are involved, this process becomes extremely complex. In such conditions, the challenge is not only selecting the most appropriate action, but rather doing so when the available information is incomplete or imprecise. This paper investigates how degrees of uncertainty and risk influence decision-making behaviour, by employing decision-making models like the discrete Bayesian network and fuzzy logic. Classic decision-making systems are often intricate and unreliable when handling uncertainty and risk. The discrete Bayesian network model exhibits a more precautionary or risk-averse behaviour when recommending actions, due to its elevated number of evacuation recommendations, whereas fuzzy logic shows a highly permissive behaviour due to its tendency of proposing warnings frequently, leaning towards precaution rather than full action. Given the lack of ground-truth data to evaluate accuracy, this investigation looks at qualitative results on how each model interprets uncertain and risk information, and how these interpretations impact the two AI models' decision-making outcomes in different ways.

Keywords: Decision-making · Uncertainty · Risk · Extreme weather events · Flood action recommendation · discrete Bayesian network (DBN) · fuzzy logic (FL)

1 Introduction

Decision-making is a crucial consideration in our daily life, involving both conscious and unconscious actions. Although definitions vary based on context, the central message conveyed remains consistent across domains, with principles that are kept unchanged. This thesis examines flood risk assessment in extreme weather, focusing on decision-making under uncertainty and risk, crucial in mitigating the increasing threat posed by flood levels caused by climate change.

The concept of decision-making is interpreted differently across fields. Some researchers interpret it as the chosen act between two or more courses of action

[2]. In the discipline of artificial agents, “decision-making discusses various ways in which actions can be selected from multiple internal states (beliefs, desires, and intentions)” [3]. Amongst these definitions, decision-making will be referred to as the process whereby an individual, group, or organization reaches conclusions about what future actions to pursue, given a set of objectives and limits on available resources [1]. This definition is especially relevant within the context of extreme weather assessment. In flooding areas of concern, authorities are called upon to decide whether to issue evacuation warnings even with limited and uncertain evidence.

For this study, decision-making is a critical factor with high repercussions, notably in situations where ground-truth labels are not available. Hence, the thesis investigates how AI models behave when recommending actions under uncertainty and risk, rather than focusing on enhancing accuracy prediction. Notice that we are not comparing quantitative results when employing *discrete Bayesian networks* and *fuzzy logic* for decision-making, but rather qualitatively analysing how these two strategies manage uncertain information and affect the decision-making procedure.

Extreme weather events increase decision-making complexity due to *uncertain* outcomes, *high-risk* forecasting challenges, and pressure on timely response measures. These decision-making exposures can be classified as innate effects that arise due to the repercussions of each action. Reinout Heijungs emphasizes that uncertainty refers to any situation in which we lack any precision, definite, or complete information [4], whereas for risk, Charles Yoe reports it as a measure of probability and consequence of uncertain future events [6]. It is noteworthy that (*uncertainty* $\neq$ *probability*) since *probability* is described by Dr. V Geetha et al. as a mathematical concept that presents the likelihood of an event taking place [5], aiding in expressing the likelihood of an outcome based on possessed information. Unlike probability, uncertainty is an intrinsic feature of information and must be represented. Interestingly, it is expressed numerically between 0 and 1, where 0 is an inconceivable event, and 1 shows a likely event.

The complication arises when these elements (i.e., risk and uncertainty) interact, changing their relative influence level, and bringing about critical consequences. For this reason, this brings about the following research questions:

To what degree can a decision be justified within flood risk assessment, particularly in scenarios of high risk and high uncertainty? How do various decision-making models, such as; discrete Bayesian network and fuzzy logic, handle uncertainty and risk in such cases, and how does these interpretations affect the advised action?

This paper intends to explore how a couple of decision-making techniques are administered under the influence of two noticeable properties: uncertainty and risk. As the paper progresses, the target will narrow down towards observing uncertainty and risk, remodelling decision-making outcomes produced by each technique in extreme weather events. Addressing the problem will require a complete understanding of decision-making, uncertainty, and risk individually,

along with, comprehending their relation in open environments. Lastly, this paper aims to explore and analyse how discrete Bayesian networks and fuzzy logic handle uncertain information and risky situations and how this further influences decision-making in the context of flood risk assessment.

2 Background

This background section explains in detail the understanding of key concepts such as decision-making, risk, and uncertainty in distinct domains, but deeply in the AI environment, along with some mathematical explanations implemented in the coding notebook. Additionally, to close this section on a high note, a subsection associating the keywords, describes the direct effects amongst these is significant and highly noticeable.

2.1 Decision-making

Decision-making is a natural act that occurs both consciously and unconsciously when an individual has full information over multiple fields. In artificial agents, systems like Autonomous Vehicles (AVs) are compelled by algorithms without any sign of consciousness.

It should be emphasised that quantifying decisions supports reasoning and leads to more robust decision outcomes, while alleviating the negative effects that uncertainty can bear on decisions. Furthermore, decision-makers are shifting towards more AI approaches that modify uncertainty in dynamic contexts, besides only implementing conventional techniques such as expected value theory, decision matrices, expected utility, and value of information [32].

Pavla Rotterová et al. describe decision matrices as an essential modelling tool for decision-making problems under the influence of risk. They depict it as an evaluation process by which numerous outputs are evaluated based on risk before making a satisfactory choice [7]. Decision matrices make decision-making challenging by presenting events horizontally, actions vertically, and probabilistic and uncertainty values in rows and columns. In actuality, most decisions do not yield specific events, so as decision-makers, it becomes necessary to designate potential outcomes as having a probability. As mentioned earlier in the introduction, probability and uncertainty are not synonyms. However, probability can model uncertainty, increasing its practicality for representing the likelihood and certainty of an event occurring. This makes probability based decision-making approaches, such as, expected utility theory, useful due to its ability to quantify decision-making choices through the use of payoffs. By quantifying decisions, a systematic framework can be provided, aiding in the evaluation of options more objectively. A payoff is the outcome received from each event possible within a given scenario [9]. Payoffs represent the consequences of actions in the form of positive or negative outcomes. With the combination of probabilities and payoffs, the decision matrix assists in providing logical structure to decisions, which

enables decision-makers to optimize their decisions. The decision matrix also offers a structured approach to managing information.

Consequently, from the classical decision theory, payoffs and probabilities are attained, which can be employed in calculating the expected value [33]. In Shu Li’s view, “expected value (EV) [1] [...] is the sum of its payoffs weighted by their probabilities” [10]. Exhibited in the notation $E[X]$, which represents the expected value of the random variable X . EV consists of the sum of all possible values of x in X , and the multiplication between the corresponding probability ($p(x)$) and the value x itself, representing the values of the random variable X . The formula of expected utility is:

$$E[X] = \sum_{\forall x \in X} p_x(x)x \quad (1)$$

On the contrary to expected values $E[X]$, expected utility [2] presents “making rational choices when they are not sure of which outcome will result from their actions. That is, people’s behaviour under uncertainty based on their perception” [11]. In this scenario, the equation follows a similar notations to the expected value. Using the Stanford Encyclopedia, the expected utility is shown as the sum of the probability of k multiplied by the utility of jk .

$$U_j = \sum_{\forall k} p_k u_{jk}(\forall j) \quad (2)$$

These results can then be analysed by the decision maker, ensuring that only the act with the highest expected utility is selected. This maximisation of the expected value hypothesis, indicates the act a^* with the highest U_j . Referring to Equation 3, it is intended to find the j that leads to the maximum (*argmax*) expected utility value.

$$a^* = \underset{a}{\operatorname{argmax}} U_j \quad (3)$$

Finally, the last approach within decision-making is the value of information. According to Scylla and Charybdis, value of information (VOI) analysis provides methods for optimising efforts and resources to gather, process and apply information to help decision-makers achieve their objectives [17]. In such a manner, it indicates the value of information or data that is pertinent to making a sophisticated decision.

2.2 Uncertainty

Having established the theoretical foundations of decision-making and its approaches, this section intends to examine the background of uncertainty and how AI models handle uncertainty within decision-making. Uncertainty can be described in different ways in correspondence to the domain it is being referred to. Uncertainty is categorised into two segments: aleatoric and epistemic. Eyke Hüllermeier et al. take the view that aleatoric uncertainty refers to the notion of randomness, the variability in the outcome of an experiment which is due to inherently random effects, whereas epistemic uncertainty is referred to as uncertainty caused by a lack of knowledge [21]. Nonetheless, to keep this succinct, Peter Byrne’s definition will be used, where he defines uncertainty as anything

that is not known about the outcome of a venture when the decision is made [18].

In this regard, while conventional calculative decision-making exists (section 3.1) to tackle such uncertainties, there are also AI methods that can be extensively used to specifically focus on making decisions and handling uncertainty in various ways. The two AI models selected to address epistemic uncertainty are discrete Bayesian networks (DBN) and fuzzy logic (FL). A discrete Bayesian network (DBN) is a probabilistic graphical model for representing knowledge about an uncertain domain where each node corresponds to a random variable and each edge represents the conditional probability for the corresponding random variables [19]. Example can be found in section 3.2. Events in DBN graphs hold probabilities assigned a priori and through new knowledge, which can be updated through new information gathered, and can change the probability of an (x) event occurring, possibly increasing or decreasing the likelihood value. Anthony C. Constantinou argued that DBNs are widely recognised as the most appropriate and beneficial method for modelling uncertainty in areas with limited data as a means to avert overfitting [13]. Likewise, this AI model struggles to learn parameters in situations where “training data is incomplete or sparse or when multiple hidden nodes exist” [14]. Moreover, DBNs require precise numerical values to quantify probabilities, leading to more error-prone and less natural decisions made in everyday situations. With the use of fuzzy logic, it is possible to solve these issues by “provid[ing] a tool for representing uncertainty by permitting an input to have a value in a range of values between 0 (completely false) and 1 (completely true)” [22]. Fuzzy logic (FL) is a multivalued logic, that allows intermediate values to be defined between conventional evaluations [20]. The fuzzy logic technique implemented in this thesis is the Mamdani system. It involves several steps like: defining linguistic variable, constructing membership functions, applying IF-THEN rules, developing inference engine, and crisping output data through the usage of defuzzification. This is achieved by importing the *skfuzzy* Python library.

Fuzzy logic can be of great help, resembling the way humans make decisions while handling vague or incomplete data. Regarding its ability to handle uncertain values, it can treat uncertainty and vagueness adequately. Such vantage is perceptible in step (2) of the step-by-step fuzzy procedure introduced below:

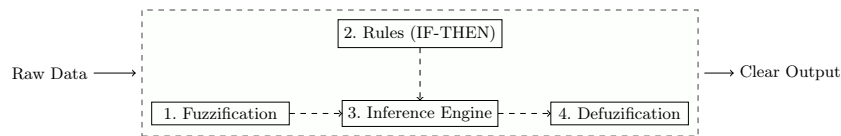


Fig. 1: fuzzy logic procedure representation

As a consequence of the system not taking into account strict yes/no answers, and instead recognising raw data introduced into the system as more than one

category, fuzzy logic’s disadvantage arises regarding subjectivity. This is because, fuzzy sets and rules established by experts can potentially introduce bias [16].

By way of illustration, within extreme weather scenarios, various models can be accounted for and incorporated to aid decision-making under distinguishable uncertainties and risks. In the case of a “flood”, *discrete Bayesian networks* could be an adequate model when several factors are involved and affect the likelihood of a flood developing. Based on the repercussion level of each element upon flood probability, the odds of this arising are higher or lower. On the other hand, *fuzzy logic* provides a more rapid and human-like approach which helps with decision-making, removing the conditions of issues generated from vague inputs. Fuzzy logic first fuzzifies the quantitative variables. These numerical inputs can be grouped into three major groups (Eg, “Low”, “Medium”, or “High”). Once the input data is in the indicated format, IF-THEN rules are implemented (*IF Deforestation is Medium AND Siltation is Medium THEN FloodProbability is Medium*) and used by the inference engines to compute the degree of flood occurrence. The last step is defuzzification, turns these fuzzy values into a crisp decision value.

2.3 Risk

Related to decision-making, the other innate property that counterbalances uncertainty after a decision is risk. Before addressing the main idea more profoundly, it is essential to define the difference between uncertainty and risk. While uncertainty refers to as anything that is not known about the outcome when the decision is made [18], risk, described by S.H. Begg et al., is a particular kind of uncertain event (or statement) — an event which itself, or its consequence, is considered “negative” to someone in a given context [23]. Risk can be calculated beforehand, and is often formulated through equations, making it an accepted approach in AI decision-making and engineering domains [4]. This equation not only conveys how certain/uncertain the information possessed is, but also presents how dangerous it would be to take a decision with a high risk. The equations in question are:

$$\text{Risk} = \text{Uncertainty} + \text{Damage} \quad (4)$$

$$\text{Risk} = \text{Uncertainty} \times \text{Damage} \quad (5)$$

Despite both equations not being very practical and yielding misleading results, especially when comparing a low-probability high-damage scenario with a high-probability low-damage scenario [24], they are considered of great use when confronting expected value calculations.

Another alternative to address risk is presenting the systematic framework proposed by ISO 31000. ISO defines risk as a “risk management process”, divided into three sections [25].

- risk identification — aimed at finding, recognising, and describing risks;
- risk analysis — studying the uncertainties, risk sources, consequences, likelihood, events, scenarios, controls, and their effectiveness;

- risk evaluation — involves comparing the results of the risk analysis with the established risk criteria.

Ensuring a systematic and robust approach in managing uncertainty and risk in AI decision-making.

2.4 Interplay between Decision-making, Uncertainty and Risk

As demonstrated in the above sections (2.1-2.3), decision-making not only consists of displaying the most appropriate decision of interest, but also involves accounting for the influence that uncertainty and risk can exert. These two intrinsic factors introduce additional layers of complexity that must be considered. Hence, this constitutes a structured process across several domains, in which various combinations of decision-making techniques and uncertainty-aware AI approaches are used to mitigate possible after-effects of suggested definite decisions.

Given the complexities of decisions, regarding uncertainty and risk, in particular extreme weather occurrences like floods, implementing strategies to address these consequences would be a viable option. This resolution occurs when uncertainty emerges from the incompleteness and noise of the input data. Therefore, assigning numerical values can lead to logical judgments. Quantifying these uncertainties through the use of stochastic processes and probabilistic models, such as expected values, decision trees, decision matrices, Markov chains, and discrete Bayesian networks, can bring about more trustworthy and robust risk evaluations. Two approaches supporting reasoning and inference under uncertainty are discrete Bayesian networks and fuzzy logic, both further explained in the following passages. Remarkably, risk is generally present, unlike uncertainty. Risk can frequently be represented with known probabilities or expressed in a measurable way (for example, as “Low”, “Medium”, or “High”), while uncertainty is based on unknown or undefined probabilities.

3 Methods

The dataset used in this project is “flood.csv”, which originates from the Kaggle database. It includes many features relating to flood risk assessment and action recommendation, as well as, environmental and social characteristics, and contains 50000 rows and 21 columns. At first, a dataset exploration was executed along multiple sites (“Google Dataset Search”, “Kaggle”, “Data.gov”, and “Earth Data”) in search of a dataset that could potentially satisfy the following bullet points:

- Main topic domain ("Decision-making under extreme weather risk and uncertainty")
- Balanced Classes
- Clear target variable

– Complete Feature set

The dataset was chosen for its simplicity and broader coverage, overcoming imbalances found in other extreme weather datasets. It was also chosen due to its completeness, unlike previous ones selected which contained incomplete cells (N/A). The current dataset has no missing numeric values in its columns, and each column is of int64 type, making it a suitable dataset for AI reasoning models with minimal and expeditious preparation.

The important dataset features that ease the insight into flood probability and decision-making under the effect of environmental and human factors were only considered [26]. However, only seven features were taken into account, making it evident that only the environmental conditions could affect the flood decision. These are the seven columns implemented [26]:

1. MonsoonIntensity: Higher volumes of rain during monsoons increase the probability of floods.
2. TopographyDrainage: Efficient drainage helps drain rainwater and reduce risk of floods.
3. RiverManagement: Proper river management, can improve water flow and reduce floods.
4. Deforestation: Deforestation reduces soil's ability to absorb water, increasing the risk of floods.
5. DrainageSystems: Well-maintained drainage systems helps drain rainwater and reduce the risk of floods.
6. Siltation: Accumulation of sediments increases the risk of floods.
7. FloodProbability: The overall probability of flooding in the region. This is the target Variable.

3.1 Preprocessing Steps

In the first cell, all imports were entered manually to ensure that each task in the coding section could be executed successfully. The next step involves loading and reading the Kaggle dataset using `data = pd.read_csv('ExtremeWeather-Temperature Prediction/flood.csv')`. Fortunately, the chosen dataset did not require any manipulation in any way that could alter the experiment, such as filling missing data gaps, or eliminating redundant values, etc. The only modification made to the data was discretisation.

A discrete Bayesian network (DBN) was employed instead of a continuous Bayesian network, even though the goal variable "FloodProbability" is continuous. After all, features in the data were ordinal, which meant that the discrete BN could handle them. Using `pd.qcut()`, features could be discretised into ("High", "Medium" and "Low") categories, which creates quantile bins using the number of samples in the dataset and divided them into the three equally sized categories mentioned above. This step is included to help avoid class imbalance and to make CPDs interpretable, as explained in a subsequent section.

Lastly, for data processing, the dataset was divided using `train_test_split()`, with 80% for training, and 20% for testing, with a random state of 42. This guarantees the experiment is reproducible and simplifies debugging the problem while comparing results.

Conversely, the preprocessing step in fuzzy logic was simpler, compared to the discrete Bayesian network. The only shared step between both methods was loading the data and reading it into a pandas DataFrame. Simplicity arose from the avoidance of discretisation and the creation of fixed boundaries for probability tables, which was not a requirement in fuzzy logic. Instead, it defines fuzzy logic variables using *skfuzzy* and handles continuous inputs using membership functions *fuzz.trimf*.

3.2 Model Structure

For Discrete Bayesian network, the network structure was illustrated to increase the interpretability of the environmental factors that could influence “Flood-Probability” directly or indirectly.



Fig. 2: Model Structure

As shown in Figure 2, two images have been incorporated. The image on the left-hand side (LHS) provides an overview of each node represented as the variables involved in the flood decision. Simultaneously, the arrows/edges in-between the nodes indicate the relationship between parent-child nodes (e.g. “DrainageSystems” is the parent node of “FloodProbability”; meaning it directly alters the target variable “FloodProbability”).

Meanwhile, the image on the right-hand side (RHS) represents the code from which the visualisation originates. In this scenario, factors that directly impact the target variable are “MonsoonIntensity” “Urbanization”, “RiverManagement”, “Siltation”, “Topography” “Drainage”, and “Drainage Systems”. Whereas the only factor to have an indirect impact is “Deforestation”. Deforestation makes it harder for soil to absorb water [27], causing the soil and reservoir to end up in the rivers (siltation). If both levels (deforestation and siltation) are high, “Flood-Probability” will inevitably be high. Hence, these pair of nodes were structured this way.

The model structure of fuzzy logic was also extremely similar. This was executed by manually defining all antecedents (“MonsoonIntensity,” “DrainageSystems”, “RiverManagement”, “Deforestation”, “Urbanization”, “TopographyDrainage”,

and “Siltation”) and the consequent (“FloodProbability”), aiming to set up a coherent fuzzy logic system, where the antecedents are the inputs, and the consequents are the output. The system controls the outcome based on the values of varying inputs acquired from the flood.csv dataset.

Following this, the membership functions were used to specify the margins of the fuzzy sets (“Low”, “Medium”, and “High”), while IF-THEN rules were used to automate the decision-making process for flood probability based on environmental factors.

While FloodProbability is the ultimate variable for action recommendation, both DBN and FL reason using environmental features in determining or interpreting FloodProbability which, in essence, makes the decision context-sensitive instead of fixed. Hence, FloodProbability is not an independent input, but instead is influenced by direct or indirect related environmental factors. In consequence, it contributes towards the final environmental decision, rather than acting as an independent factor in either model to directly recommend an action on its own.

3.3 Parameter Learning

Once the structure of the discrete Bayesian network is defined, the upcoming step involves learning conditional probability distributions (CPDs). CPDs are a tabular way of discovering the likelihood of the child node occurring given all possible combinations between the categories of the parent nodes in the network. For instance, if we have “MonsoonIntensity” and “DrainageSystems” labelled as parent nodes of “FloodProbability”, we would question: *What is the likelihood of a flood occurring based on the combination of monsoon intensity and the quality of drainage systems.* In mathematical terms, this counts on both *conditional probabilities* $P(A|B) = P(A \cap B)/P(B)$ and *Bayes’ Rule* $P(C|D) = P(D|C) * P(C)/P(D)$. Conditional probability is as simple as the “likelihood of an event or outcome occurring based on the occurrence of a previous event or outcome” [28], represented as $P(A|B)$, which indicates the probability of (A) occurring given (B). For this specific case, this can be expressed as $P(\text{FloodProbability} = \text{High} | \text{MonsoonIntensity} = \text{High}, \text{DrainageSystems} = \text{High})$, indicating the probability of “FloodProbability” being “High”, given both “MonsoonIntensity” and “DrainageSystems” being “High”. In contrast, Bayes’ Rule is known as a “rigorous method for interpreting evidence in the context of previous experience or knowledge” [29]. Identifying the probability of the causes given the effect and updating the discrete Bayesian network for decision-making.

Two well-known parameter estimation techniques were utilised to ensure a panoramic experiment; maximum likelihood estimation (MLE) [7] and Bayesian estimation [8]. While the similarities highly involve their reliance on the observed data, they only differ in the way they handle prior knowledge. Both methods use likelihood function, which rises to the occasion when trying to find the possibility of data and its parameters. In turn, the “random variable X that has a probability distribution $f_X(x)$ with probability density function (pdf) $f_X(x)$ parameterized by a parameter θ ” [30], illustrating the likelihood function (given θ) just as shown below:

$$L(\theta; x) = f_X(x | \theta) \quad (6)$$

The distinction between the two methods is fairly straightforward. The MLE function, as the name suggests, is capable of finding the parameters that lead to the maximum likelihood function. Referring to Equation 7, representing MLE, we can observe that it “can be derived as follows. Let there be a random variable whose pdf, $f_X(x)$, is parameterised by a parameter θ , [...] where $f_X(x|\theta)$ is the pdf of X for a given θ ” [30]. Additionally, MLE estimates Conditional Probability Distributions (CPDs) by calculating the relative frequencies from the dataset. However, it has a major disadvantage, it can overfit if the data is sparse. To overcome this issue, we can introduce Bayesian estimation. The core idea behind this technique is to start with existing prior CPDs before the data has already been observed, and previous knowledge of the distribution has been settled. In this context, we refer to prior belief as the belief about something before diving into the data. Bayesian estimation commences with assumptions (Prior Conditional Probability Distributions) about the probability distribution of the data without previously observing it; these are represented as pseudo-counts. These prior pseudo-counts are continuously updated by merging them with the actual observed counts, which are acquired by counting the frequency of the variables. The combined counts are later normalised to produce updated CPDs. As shown by Leonardo Leoni et al., the Bayesian estimation Equation 8 is made up of the likelihood $f(x|\theta)$ multiplied by prior distributions $\pi_0(\theta)$, divided by the integral of itself (likelihood multiplied by prior distributions) with respect to θ [31].

$$\hat{\theta}_{MLE}(x) = \text{Arg max}_{\theta} \{L(x; \theta)\} \quad (7)$$

$$\pi_1(\theta|x) = \frac{f(x|\theta) \pi_0(\theta)}{\int_{\theta} f(x|\theta) \pi_0(\theta) d\theta} \quad (8)$$

3.4 Inferencing Method

In discrete Bayesian networks, inference techniques include answering probability queries and calculating probabilities across the network, given the other variables. This model is useful in understanding various conditions of likelihood, and making some reasonably valid inferences, which can help in making decisions. Referring to the coding notebook, the inference technique included is *variable elimination*, aiming to only show the factors that affect the likelihood of a flood. This algorithm start by calculating the probabilities of all the variables. In due course, variables were removed successively, until left with those that were considered to be critical towards the target variable. This way, probability calculations are narrowed and simplified so only the necessary variables are included, resulting in an elementary depiction. Later, it computes conditional probabilities using the Joint distribution, where the inapplicable variables are reduced and marginalised.

The inferencing method in fuzzy logic is handled by both the `ControlSystem` and `ControlSystemSimulation` functions that belong to the `control` import. Concisely, `ControlSystem` assembles all the 18 rules defined in the code, indicating how the model should behave based on the input conditions. Alternatively, `ControlSystemSimulation` initializes the fuzzy control system based on the rules defined and the data provided.

In the fuzzy logic approach, action recommendation consists of two steps. First, the crisp output from defuzzification is mapped to a qualitative risk level based on a threshold. If the result is greater than 0.35, risk level is classified as "High", if it is between 0.2 and 0.35 the risk level is "Medium", otherwise it's "Low". Next, a utility matrix is used to evaluate the cost/benefit of each action ("Evacuate!", "Warning", or "No Issue") given the risk level. The action with the highest expected utility is selected. The purpose of this design is to allow these AI models to balance taking risks with being prudent, and making decisions based on how harmful the flood is expected to be.

3.5 Model Validation

The purpose of model validation is to demonstrate and evaluate how well the model supports decision-making and its recommendations ("Evacuate!", "Warning", and "No Issue") under the effect of uncertainty and risk. As mentioned in Section 3.1, the `traintest_split` was performed as follows (80% training and 20% testing) on the original dataset. From that, validation was conducted on 20% of the test set. The validation process assesses the DBN's behaviour on unseen data by splitting the dataset as mentioned earlier. The model was trained on a training set, while all decision inferences from the model were performed on the test set. Although no hyperparameter tuning was conducted, this process ensured that the model's recommendations still performed reasonably well, rather than memorising patterns. The evaluation of this approach is not a typical classification accuracy. Instead, the model validates on probability distributions ("High", "Medium", "Low"), a way to make decisions ("Evacuate!", "Warning", and "No Issue") based on expected utility calculations for action recommendations.

For the fuzzy logic validation, a *for loop* was coded with the purpose of adding thresholds that defined the action taken based on the output "FloodProbability".

4 Results

This section principally presents the results of the discrete Bayesian network and the fuzzy logic models within the context of flood risk in decision-making under uncertainty and risk. The spotlight lies on the action recommendations ("Evacuate!", "Warning", and "No Issue") based on the "FloodProbability", the implication of the variables and utility function. A small comparison of the model distinction and behaviours supporting (Section 2.2) is included.

4.1 Discretization and State Count

All data/features involved in “Flood.csv” are continuous by default. However, data was discretized and divided into three-level categories (“Low”, “Medium”, and “High”) to make it more convenient to work with the discrete Bayesian network, especially when inferencing and using utility functions for decision-making. A discrete Bayesian network was implemented over the continuous Bayesian network, because discrete Bayesian network tends to perform better than continuous in smaller and limited datasets. DBN models are also more interpretable and easier when handling with categorical inputs, benefitting future explanations for decision-making under uncertainty and risk.

As mentioned in section 3.1, the code utilised for discretisation, is `pd.qcut()`, to ensure uniform quantile bins with similar sample numbers, preventing class imbalance. On the contrary, manual bin setting using `pd.cut()` was also considered, however this increases the risk of unbalanced distribution within categories. As shown in Table 1, data is discretised using `pd.qcut()`, which automatically administered the data equally, but indicates a slight inclination towards the “Low” column for all variables, except for the target variable “FloodProbability”. In said manner, this reduced the risk of flood decision-making inferences.

	Low	Medium	High
FloodProbability	16691	17298	16011
MonsoonIntensity	22022	16140	11838
TopographyDrainage	22210	16042	11748
DrainageSystems	21969	16146	11885
Urbanization	22121	16127	11752
RiverManagement	21799	16230	11971
Deforestation	21962	16173	11865
Siltation	22107	16138	11755

Table 1: `pd.qcut()` distribution (“Low”, “Medium”, and “High”)

4.2 Action Dispersal and Risk Behaviour

From the review mentioned above, key findings emerge:

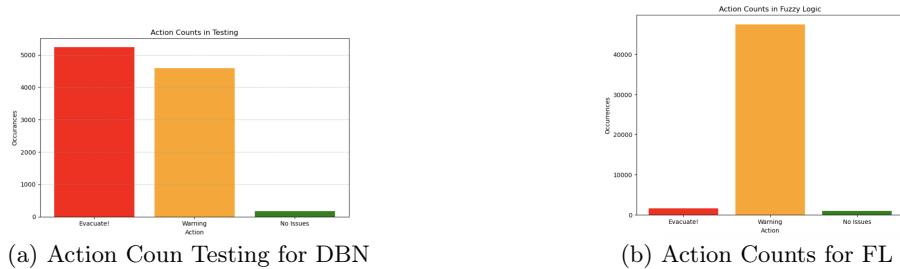


Fig. 3: Action Counts for DBN vs. FL

Concerning Figure 3(a), it is evident that the Discrete Bayesian model in testing tends to recommend the actions “Evacuate!” (5242) and “Warning” (4583) much more than “No Issue” (175). This outcome reflects that the model presents risk-averse behaviour since it tends to decide and suggest “Evacuate!” the vast majority of the time when it perceives a relatively high possibility of flood. The approach to flood warnings is cautious, aiming to avoid risky scenarios. It is often safer to evacuate, even if unwarranted, than to say “No Issue” and risk lives, *better to feel safe than sorry*. However, this introduces a high level of false alarm and a secondary risk. Repeating warnings in non-critical scenarios can diminish trust in the model and cause serious problems. A solution to risk-averse behaviour would involve adjusting the utility matrix in the coding section. The utility matrix ensures that every action taken has either a positive or negative consequence, held by a specific cost. For instance, if an excessive penalty is placed on a “No Issue” decision, in an event of an actual flood (false negative), the model is encouraged to recommend the action “Evacuate!” more often, thereby supporting safety. On the contrary, increasing the cost of false alarms, would lead to a more cautious model when regarding warnings. Altering the cost between these categories, “Low”, “Medium”, and “High”, can result in a more balanced decision distribution taken by the model, possibly striking a more trustworthy decision for civilians undergoing flood danger.

As opposed to DBN, fuzzy logic follows a different procedure on how decisions are derived from the continuous input, resulting in a different action distribution. The dissimilarity stems from how thresholds are applied for both AI algorithms. In fuzzy logic a more sophisticated threshold is applied after defuzzification, whereas for the discrete Bayesian network, a fixed margin was needed. As a result, fuzzy logic produced the following occurrences: “Evacuate!”: 1613, “Warning”: 47452, and “No Issues”: 935, as shown in Figure 3(b).

4.3 Decision-Making Under Uncertainty and Risk (Coding Process)

Both AI algorithms have an important decision-making component specifically conceived for uncertainty and risk. This is what sets this project apart from a standard classification problem. At the centre of the discrete Bayesian network implementation, it iterates over each dataset row for all variables influencing ‘FloodProbability’ and uses conditional probabilities along with the utility matrix to find expected utilities. Each variable will be evaluated, and its probabilities will be used to derive expected utilities in each case that will come from each utility matrix. As mentioned earlier in the paper, to involve risk in the equation of this procedure, by involving the utility matrix combined with the utility function, the action with the max expected utility is selected from the distinct options. Below, a code snippet is included for one of the rows:

```
Row 0 -----
Probabilities (Low/Medium/High): {'High': 0.29295407898306397, 'Low': 0.32739384567685925, 'Medium': 0.3796520753400767}
Expected Utilities: {'Evacuate!': 15.170286245785373, 'Warning': 14.744694647909158, 'No Issues': -11.373765492384607}
Recommended Action: Evacuate!
```

Fig. 4: Discrete Bayesian network (Code)

Figure 4 demonstrates the probabilities for the three categories (“High”, “Medium”, and “Low”) employed for the creation of expected utility, where the utility matrix adds the cost depending on the probability. Additionally, from this same example, based on the expected utilities, the action with the highest value is selected amongst the other, ensuring the most pertinent (“Evacuate!”, “Warning”, or “No Issue”) action is reported. This has been performed for every row of the dataset.

In the fuzzy logic approach, risk is handled in a similar manner, but applied in a different order. The utility matrix is applied after defuzzification step, based on the fixed threshold mentioned earlier in the paper. Different from DBN, uncertainty is handled differently in FL. Here, during the fuzzification process, membership functions are applied, to assign a degree of truth to the elements within the fuzzy sets, enabling the model to reason with vague input.

5 Discussion

	Evacuate!	Warning	No Issue
Discrete Bayesian network (Testing)	5242	4583	175
fuzzy logic	1613	47452	935

Table 2: Models Action Recommendation summary

As demonstrated in Table 3, both models highly indicate safety but take different actions, which could lead to similar yet different interpretations. The discrete Bayesian network exhibits a more precautionary behaviour during testing, indicating “Evacuate” 52.42% of the time. Meanwhile, the fuzzy logic model is rather more unprejudiced, presenting “Warning” predominantly (94.9%). These contrasts convey how each model deals with decision-making under uncertainty and risk differently. While DBN was evaluated and tested on 20% of the dataset (10,000 samples), fuzzy logic was performed in the entire dataset (50,000 samples). DBN’s framework includes uncertainty through probability distribution and was only evaluated on testing due to parameter learning, resulting in evacuation suggestions. In contrast, FL analyses uncertainty based on membership functions which arises from the rule-based inference, which results in more warnings. These techniques show the design of each model and guarantee suitable testing. Therefore, action recommendations under the influence of multiple variables in decision-making can vary.

The decision-making behaviours of the algorithms represent a different risk level, as mentioned earlier. DBN may be more precautionary when it comes to indicating actions due to the possible high level of risk and low level of certainty, at the cost of numerous False Alarms possibly encountered, prioritising the safety of inhabitants over the correctness or precision of the actual model when submitting an action. By contrast, fuzzy logic tends to cope with uncertainty better than DBN, stabilising the safety of the model and leading to a higher “Warning” action recommendation, displaying fewer False Alarms than DBN.

The experiment provides a new understanding of the relationship between the distinct levels of uncertainty and risk when dealing with decision-making, and how this affects the outcomes. Additionally, the data contributes to a clearer understanding of how dissimilar AI algorithms treat and process the data, leading to distinguishable distributions of action recommendations.

Due to the absence of ground truth data, results cannot confirm whether action recommendations post-decision-making are correct or accurate. As demonstrated throughout the paper, the task does not involve ordinal classification. Instead, it endeavours to take input data to assemble decisions under uncertainty and risk implementations. Additionally, the matrix implemented in both algorithms was manually built by setting higher reward actions for elevated risk situation and penalising when no action was taken. Hence, low rewards come from low risks.

Avenues for future research include possibly replacing the fixed thresholds with dynamic boundaries, which could plausibly provide an increased responsive simulation to real-world scenarios. Further, genuine flood labels allow a more sensible model evaluation of the performance. Finally, a hybrid model between FL and DBN, could enhance data interpretability and robustness of action recommendation.

6 Conclusion

The thesis aimed to explore how AI is capable of handling decision-making under the influence of uncertainty and risk by setting two contrasting AI approaches; discrete Bayesian network and fuzzy logic. Both utility-based AI models prioritised safety in high-risk and high-uncertain scenarios, exemplifying how different models justify their decisions using the same input data, and similar uncertainty and risk structure. Extending further, findings show that DBN favours a more precautionary approach, recommending "Evacuating" in instances where uncertainty was high, assuring that no risk is taken under any circumstances. On the other hand, fuzzy logic was more tolerant in situations involving high risk and uncertainty, resulting in fewer "Evacuate!" actions and more "Warning!" recommendations.

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